Design(E)-314
Lecture 3

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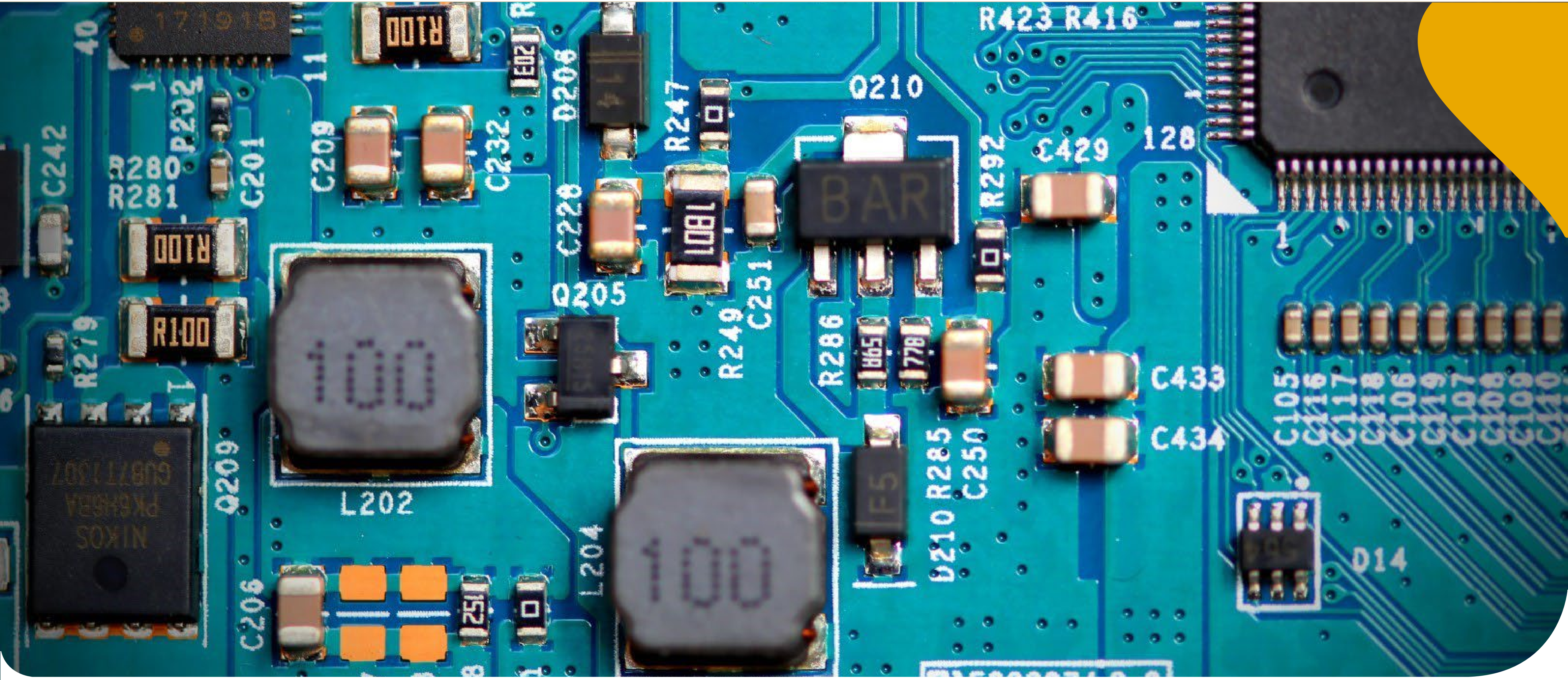
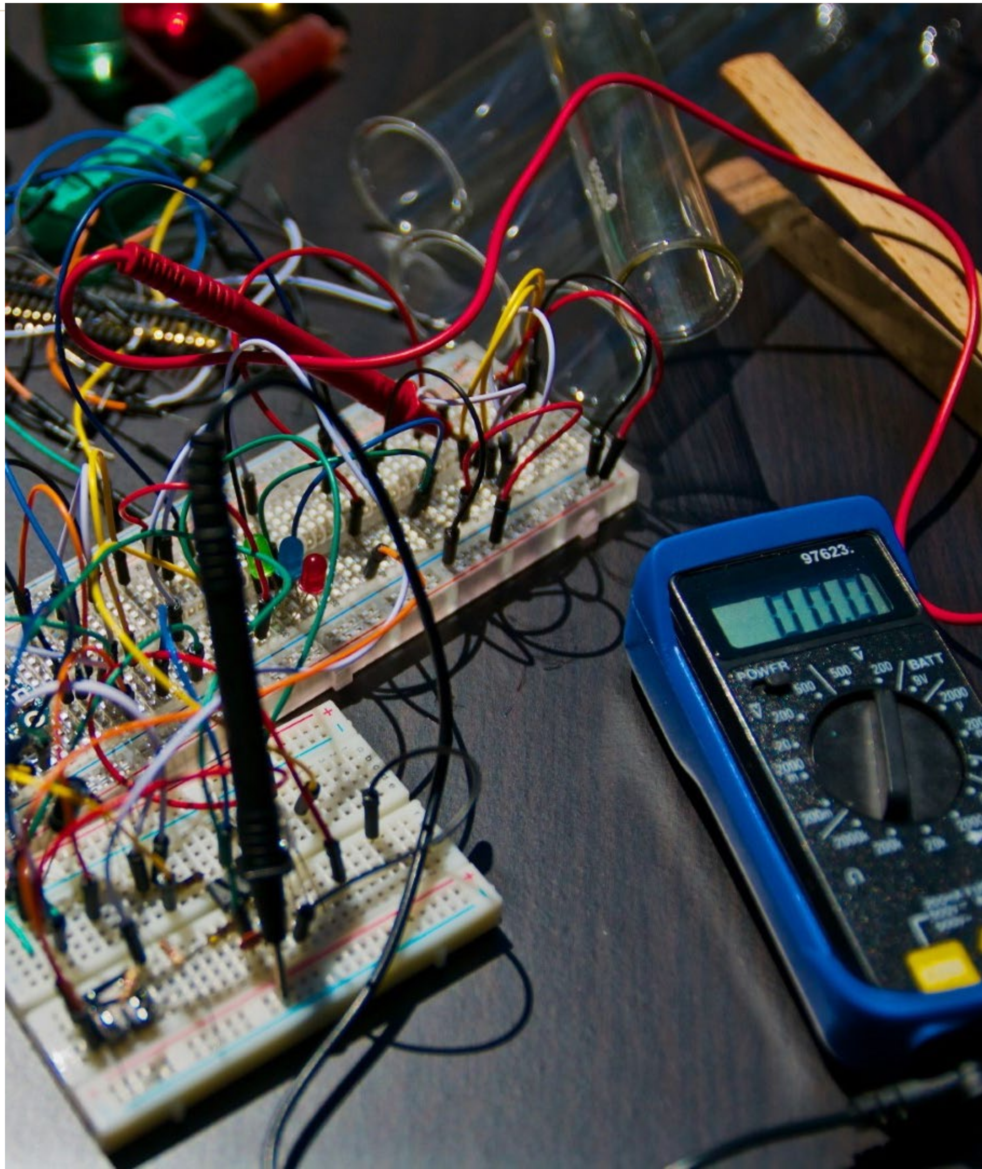


Overview

We are in the Demo 1 cycle

- Admin
- UART, TIC connections
- LMT01 (Temperature sensor)
    - Robustness & Calibration
- Live Demonstration
- Test 1
    - How to prepare, what to expect
- Ultrasonic Sensor
- Q&A



PDD - Updates

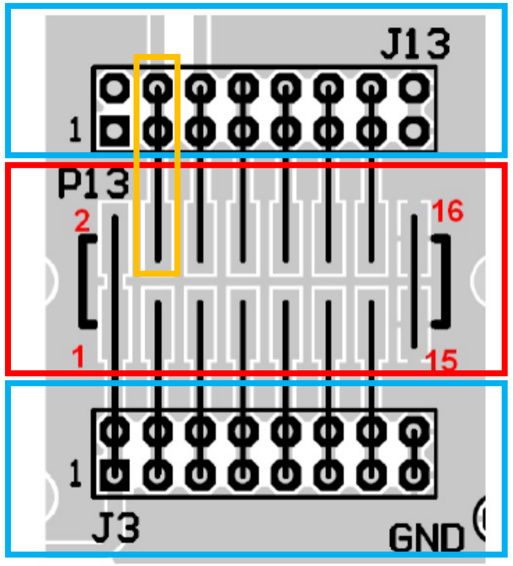
- Version 0.3 is the latest version

Version	Date	Changes
0.1	9 Feb 2026	- Initial project specification and guide
0.2	11 Feb 2026	- Added Demo 1 information in Tables 9 and 10
0.3	19 Feb 2026	- Clearly specified the Demonstration 1 UART message format for the temperature sensor in Tables 9.

- Major change in v0.3: Demo 1 – UART format for LMT01 temperature is specified

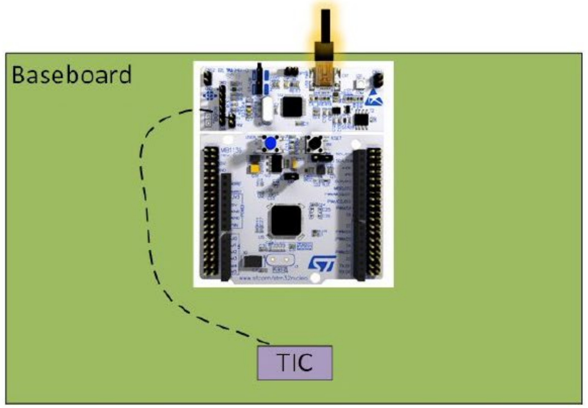
TIC Connections
General header pin connection

P13 pin	J3/J13 solder connection	Signal	Direction
1	J3 - 1,2	V_bat - 9 V Supply from test station	SB <-> TS
2	N/C	V_bat - 9 V Supply from test station	SB <-> TS
3	J3 - 3,4	UART transmit (TX from student board, RX of test station)	SB -> TS
4	J13 - 3,4	5V supply from student board (fed back to test station for verification/measurement)	SB -> TS
5	J3 - 5,6	UART receive (RX from student board, TX of test station)	SB <- TS
6	J13 - 5,6	3.3 V supply from student board (fed back to test station for verification/measurement)	SB -> TS
7	J3 - 7,8	Button UP-S1 (connected to TS for debug/verification)	SB <-> TS

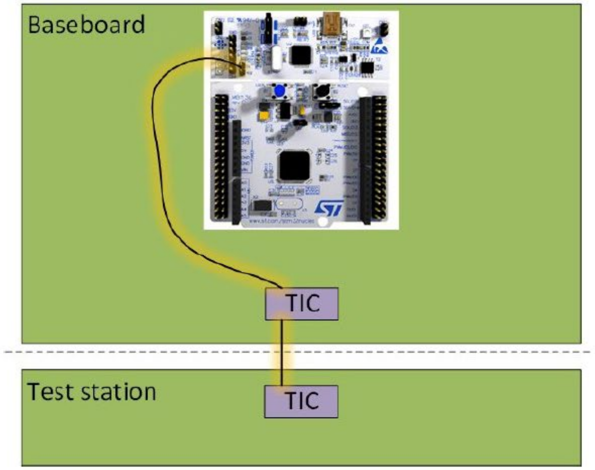


*TX and RX on the microcontroller are reversed

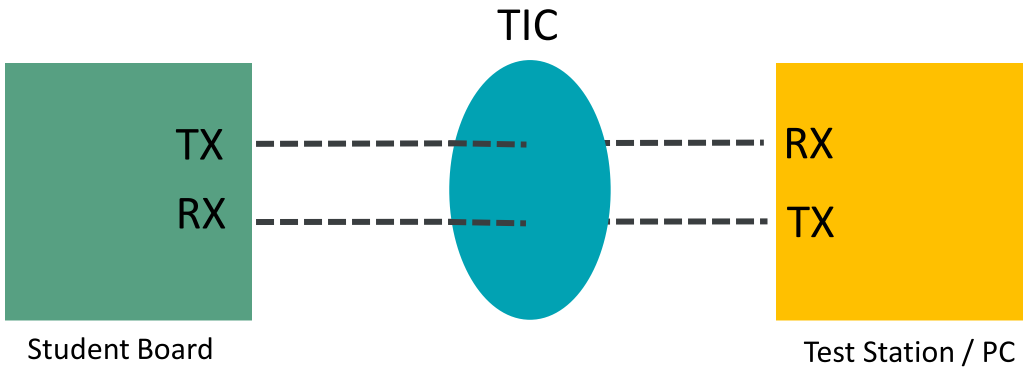
TIC Connections
UART



Student performs test or debugging without test station. UART appears as a Virtual COM port on PC



Automated testing through Test Station - UART is routed to TIC



TIC Connections
UART

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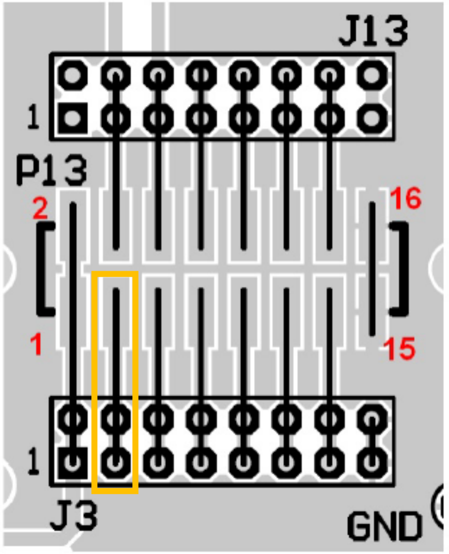

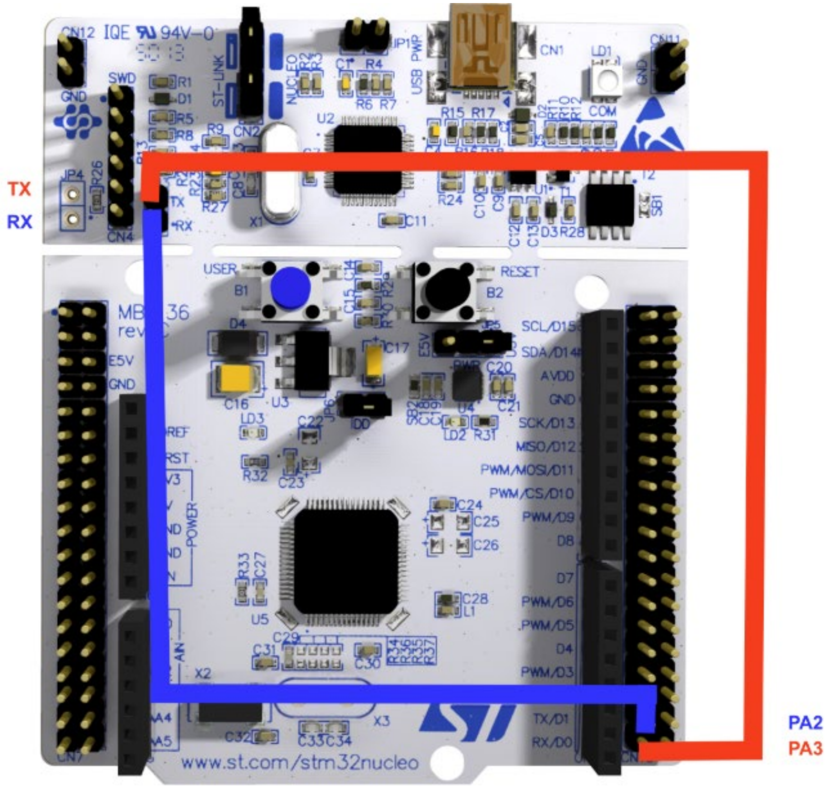

LMT01
Temperature sensor

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Implementation/Timing:

- You can use either EXTI **or** TIMER (external clock mode) to count pulses…
- If your GPIO interrupts are working…save time and also trigger an interrupt for each temperature pulse…

Pro-Tip:
To confirm if your LMT01 pulses will not overload the MCU, compare your MCU clock speed vs. your LMT01 “interrupt speed”

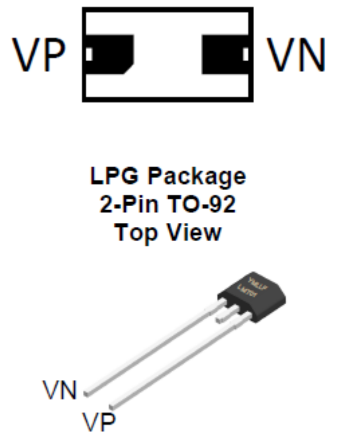
LPG Package
2-Pin TO-92
Top View

LMT01
Temperature sensor

Robustness:

- In real life, data will corrupt (now and then).
- It is your responsibility to anticipate and handle this

- Strategy: You can continuously update your temperature sensor reading in the background with the use of a circular buffer.
- Keep 5x most recent readings, drop lowest and highest values, average the remaining 3 values.



LMT01
Temperature sensor

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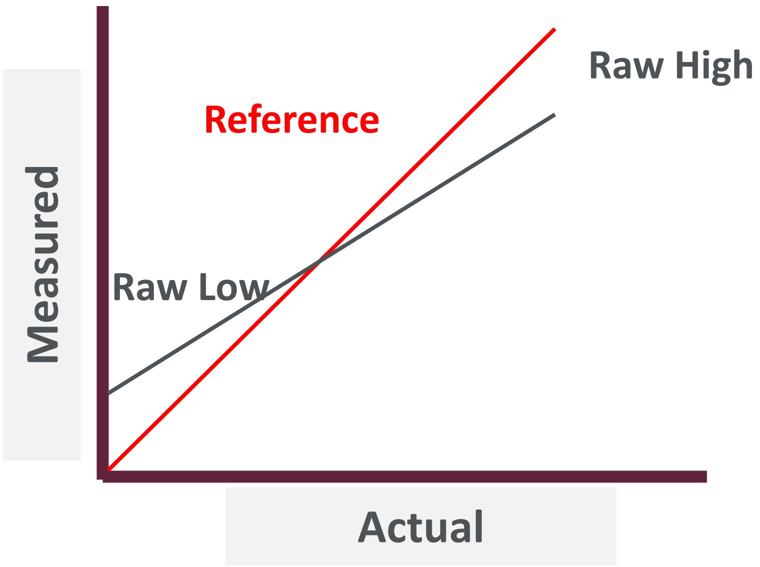
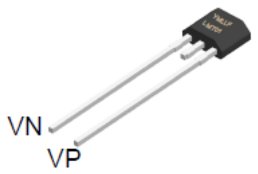
Calibration/Confirmation:

- The LMT01 is a good/expensive temperature sensor that guarantees 0.5 ˚C accuracy
- However, your code is not always "accurate"
- To ensure you are reading the right value, you must calibrate/investigate relative to a known temperature value.
- **Note**: You must calibrate relative to your anticipated range of expected temperature values
- $y = mx + b$



LPG Package
2-Pin TO-92
Top View



Demo 1 requirements
What to expect

Demo	Description	Relevant SS's
1	On power-up, the TS will expect your student number via UART using the prescribed message format, no sooner than 100ms after power-up and no later than 500ms after power-up. The TS will simulate button pushes with and without bounce for push-buttons S1-S5. The TS will override the tactile push-buttons (via TIC connector P13 pins 7, 8, 9, 10, 11) while taking photographs (of the status-indicating LEDs) to check for correct system operation. The maximum response time after button pushes is 500 ms. The TS will monitor and test the output of the digital temperature sensor (via UART). The TS will visually monitor the power status LED (D6) and status-indicating LEDs (D2-D5). The TS will monitor and test the 3.3V and 5V lines. GIT repos will be checked.	SS1, SS2 (student number), SS6 (Temperature), SS7 (LEDs), SS8 (tactile push-buttons), SS12 (5V, 3.3V, UART-RX, UART-TX).

Table 10: Demo descriptions

* Reset board to try fix 2x send

Demo 1 requirements
What to expect

SS6	*Environment Measurement:* The device must provide the ambient temperature the board is exposed to. Specifically, the device must measure: • Environment temperature using the digital temperature sensor Pushing the MIDDLE-S3 button will cause the TS to also measure the ambient temperature at the same time as the SB. The TS will compare the temperature received via UART, to the ambient temperature as measured by the TS. The format for sending the temperature value via the UART link is '@XX.X&\n' where '\n' is the newline character (ASCII character code 10, or indicated in C string escape notation as '\n')	The test is passed if the TS receives a temperature value via the UART link, and is within 2°C of the temperature value measured by the TS.

Demo 1
Live Demo

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“Live” Demo….in class

Demo 1
Time Slots

Exemptions: No exemptions

Time Slots

- Thursdays: 09:00 – 12:00 **M&M and Comp. Science**
- Fridays : 10:00 – 13:00 **E&E (all focus areas)**

Logistics

- You must demo at least once within the time slot
- -10% penalty for each additional demo attempt
- 2x Rows on demo day
  - Row 1: priority to all students who demo the first time
  - Row 2: Students who want to re-demo (must make way for students who demo first time)

Test 1

Content

- All Demo 1 requirements are subjects for the test
- Test is there to ensure that you know what you are actually doing
- Hardware, Software, Theory questions will be asked
- "Open book" i.e. bring your lecture, code & design notes to the test
  → Print out / USB
- No Google/Internet connection allowed
- We will track IP address activity to ensure this requirement

NB

Pro-Tip:
Take a screenshot of all your test answers and save it to a local word document. In case your submission fails.

Test 1

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How to prepare

- Write up on all of your hardware and software design elements that were required for Demo 1 (Content/Design unrelated to Demo 1 will not serve as test content)
- Expect: Data sheet related questions
- Expect: Hardware design questions
- Expect: Software design/code implementation questions
- Expect: Results/Interpretation questions

Pro-Tip:
Bring a pen and paper to the test

Bring a calculator to the test (much faster than PC calculator)

Test 1

Details:

- Tuesday 3 March
- Venue:
- You must write the test in your allocated lecture time slot
- If you write the test in another time slot, it will simply be regarded as an “absent” result (i.e. 0%)
    - **M&M**: 08:10 – 09:00 AM
    - **E&E, Comp. Sci**: 09:10 – 10:00 AM

Pro-Tip:
Arrive 10 minutes early, log into your PC, and complete 2-factor authentication if required.

No cellphones allowed during test

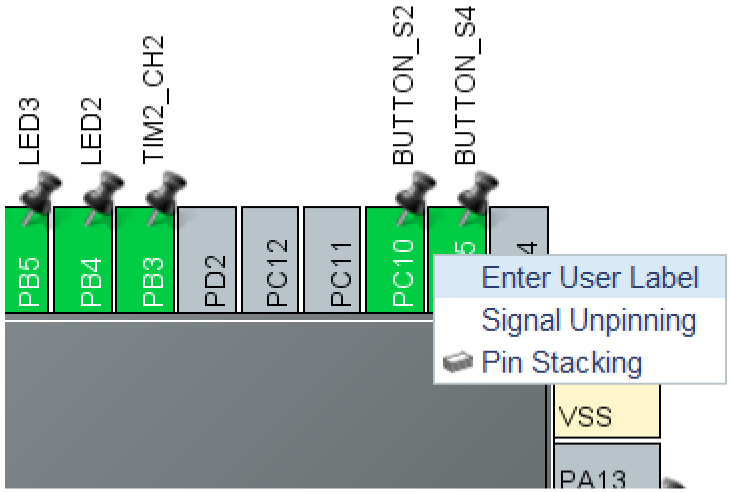
Practical 2 - Feedback
Pin config

- Did you know, you can label your pins to make your code easier?

```c
  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {

          HAL_GPIO_TogglePin(LED2_GPIO_Port,LED2_Pin);
          HAL_Delay(500);
```

```c
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
   /* Prevent unused argument(s) compilation warning */
   if(HAL_GPIO_ReadPin(BUTTON_S3_GPIO_Port, BUTTON_S3_Pin))
   {
       S3_pressed = 1; // Flag for Button S3
   }
```



GIT

- Two issues have been reported regarding the use of Gitlab:

1. Account locked / disabled

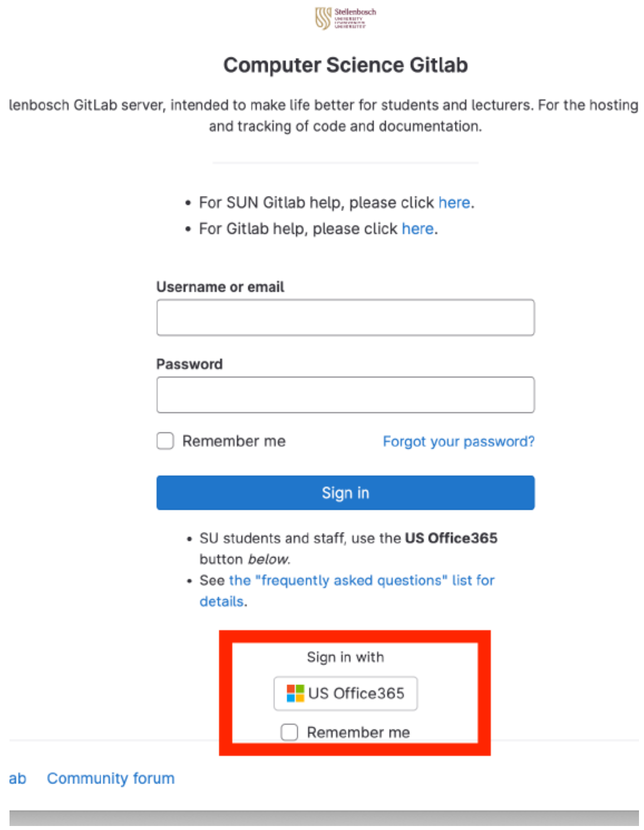
Some of you have received emails that your Gitlab account has been locked or disabled. The Gitlab administrator indicated that it is a standard security feature to prevent brute-force attacks. Your account might be locked/disabled if you use the a username/password combination instead of using the 'US Office 365' login button. The locked/disabled account is temporary and should unlock after 10 minutes. If this does not happen, please contact the Gitlab administrator (Emile Dreyer edreyer@sun.ac.za).

2. 'fatal: Authentication failed' error message when updating the remote repository by 'pushing' the local repository

The 'SU Microsoft Authentication' service using two-factor authentication (2FA). For 2FA-enabled users, a personal access token (PAT) must be used instead of a password. The '2026 - Setting up and using Gitlab' document has been updated explain how a 'Personal Access Token' can be created and used.

Why GIT?

- Over 70% of developers use Git!
- Developers can work together from anywhere in the world.
- Developers can see the full history of the project.
- Developers can revert to earlier versions of a project.
- **Note**: Sign in with US Office 365 account



HC-SR04 Ultrasonic Sensor

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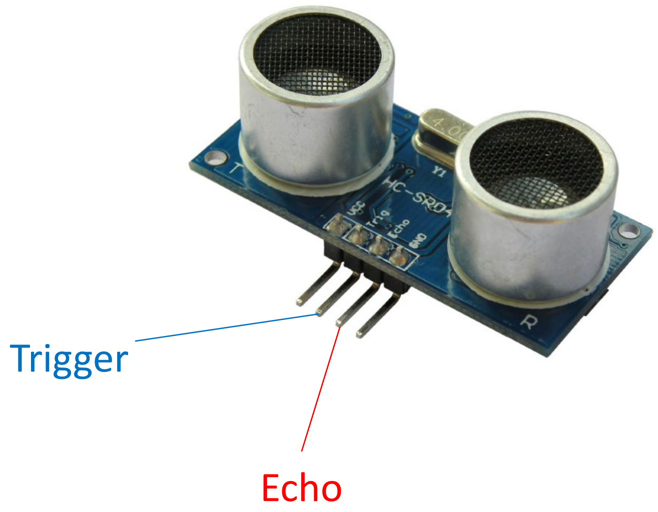
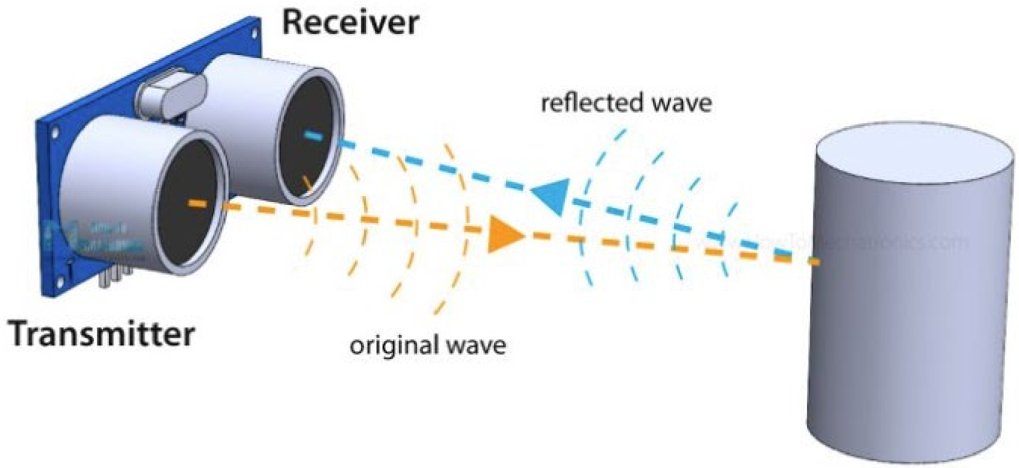
Operation

- Sensor is composed of two ultrasonic transducers
- One transmits ultrasonic sound pulses and the other receives the reflected waves
- For this project, you will use it to detect objects in near proximity to the vehicle



Parameter	Min	Typ.	Max	Unit
Operating Voltage	4.50	5.0	5.5	V
Quiescent Current	1.5	2	2.5	mA
Working Current	10	15	20	mA
Ultrasonic Frequency	-	40	-	kHz



HC-SR04 Ultrasonic Sensor

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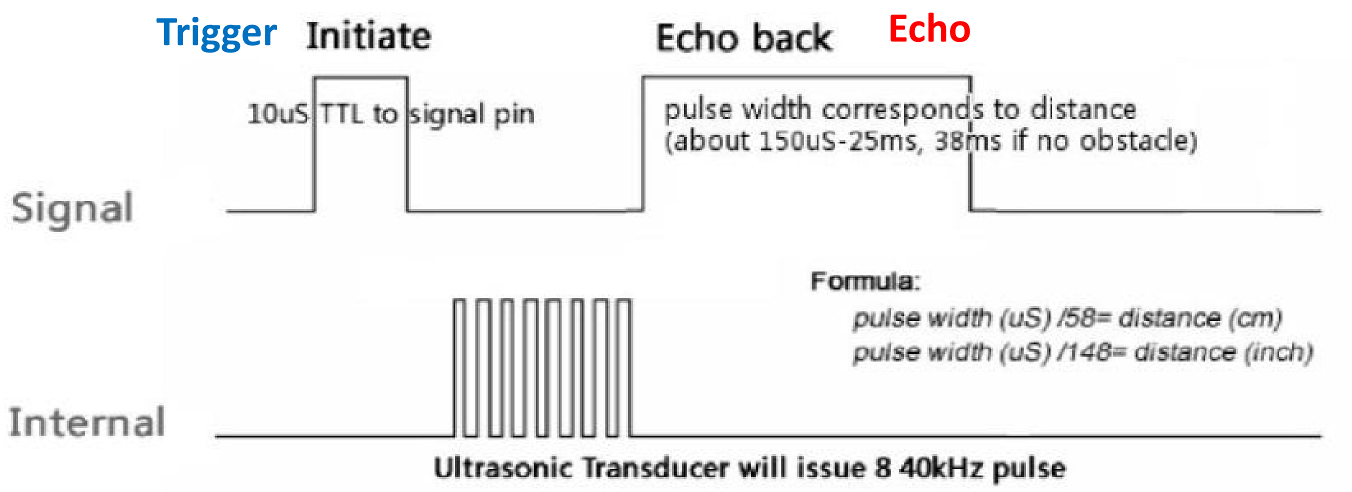
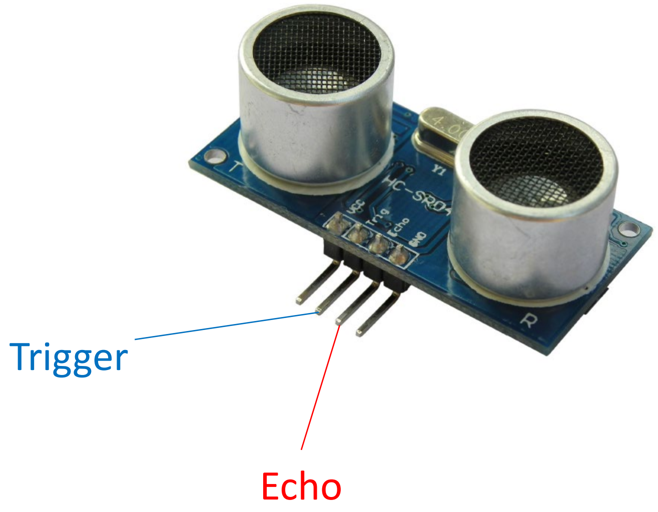
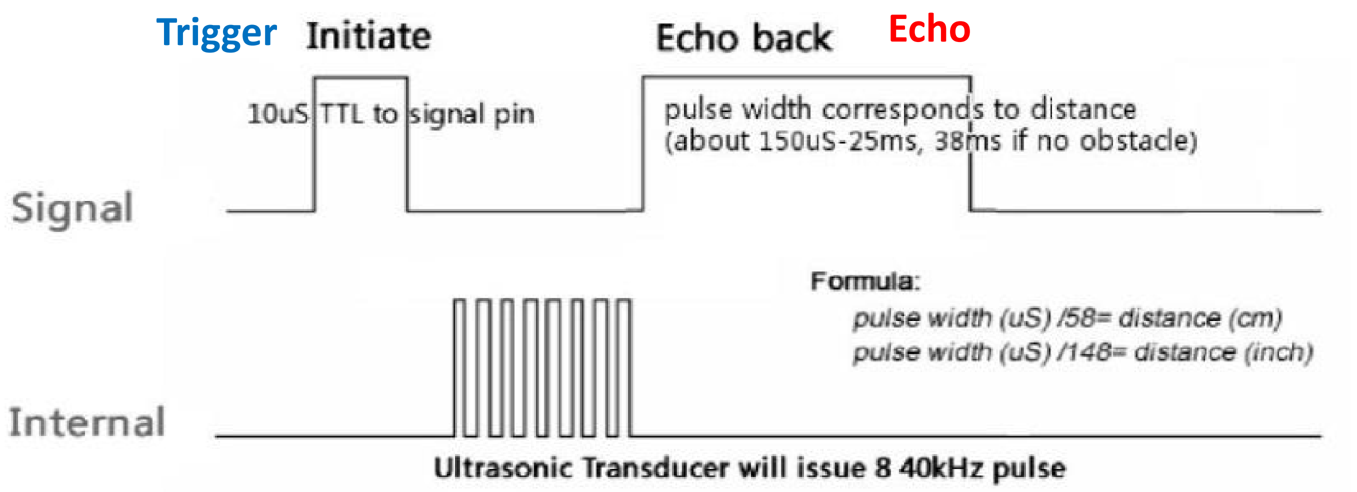
1. Start measurement with high (5V) pulse for 10us,
2. HC-SR04 activates and sends 8x cycle ultrasonic burst, of 40 kHz, which will travel at the speed of sound.
3. Echo listens for wave to be reflected from an object
4. Echo pin will time-out after 38ms (i.e. no object detected), but triggers before 38ms if object detected

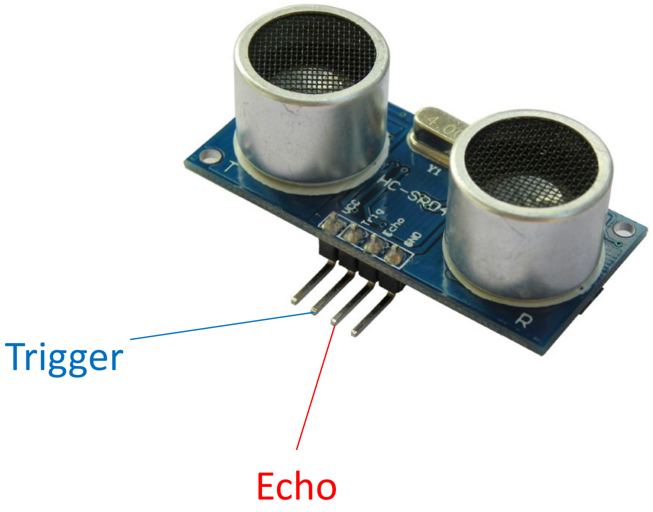




HC-SR04 Ultrasonic Sensor

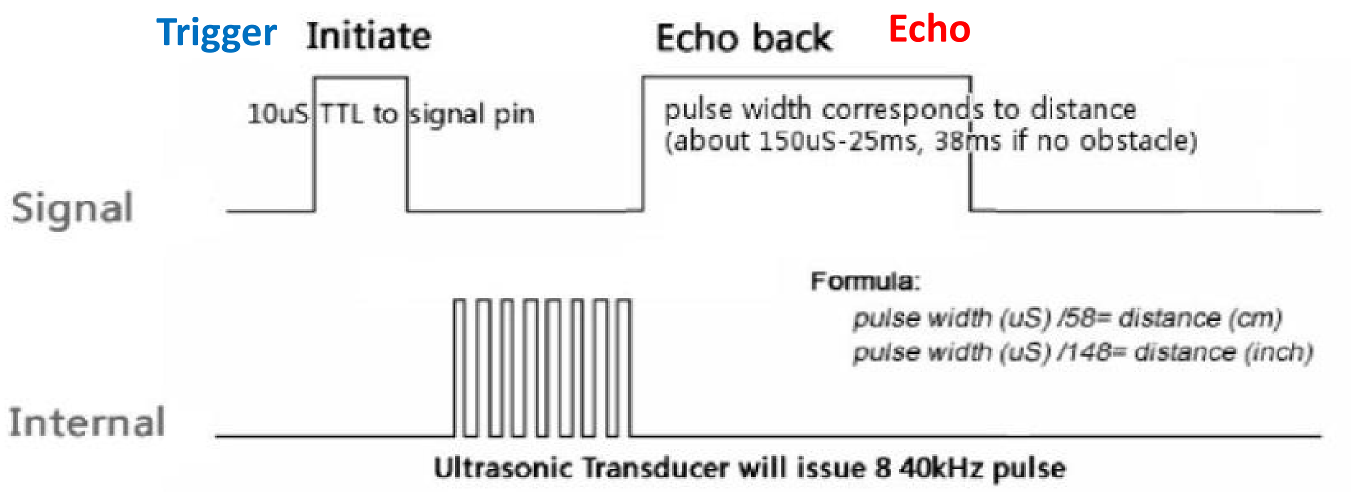
5. When the sensor detects ultrasonic sound on the receiver, it will set the Echo pin to high (5V)
6. The pulse width on the Echo pin is proportional to distance.
7. To obtain the distance, measure the width (Ton) of the Echo pin. Time = Width of Echo pulse, in uS (micro second)
8. Distance (cm) = (Speed x Time)/2
   - Speed?
   - Time?

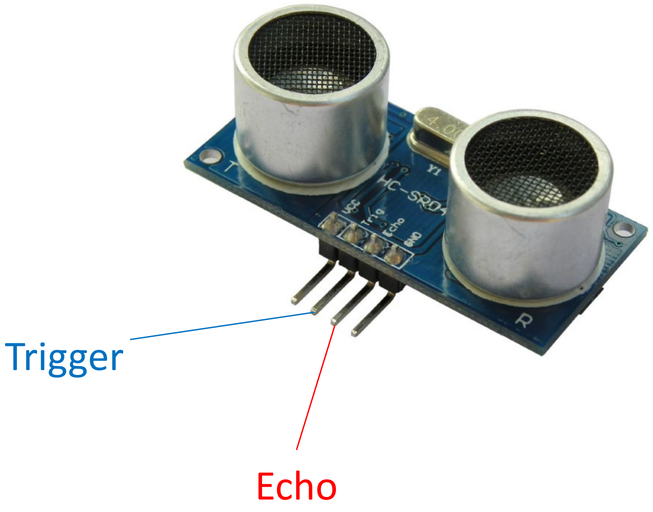




HC-SR04 Ultrasonic Sensor

- Be careful…ECHO pn is set high to 5V.
- Thus, you must manage the 5V to be 3.3V.
- Use a voltage divider circuit (k-ohms) to ensure that 5V high signal is 3.3V or less.





Questions?

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